



File System Operations

Regarding file system operations, a Web Server must:

- Read static files (HTML, CSS, images) and send them to clients
- Check if files exist before serving (return 404 if not)
- Check permissions (return 403 if forbidden)
- List directory contents (autoindex feature)
- Write uploaded files to disk (POST requests)
- Delete files (DELETE requests)
- Navigate the file system based on URL paths

Core File System Functions

`access()`

`chdir()`

`close()`

`closedir()`

`open()`

`opendir()`

`read()`

`readdir()`

`stat()`

write()

Security Critical Points

- Prevent Directory Traversal: Never serve files with "." in path
- Check permissions: Use `access()` before reading/writing
- Validate upload filenames: No "/" or "." in uploaded filenames
- Respect symlinks: Either follow carefully or reject
- Don't expose system files: Block "/etc/passwd", "~/ssh/"...